Namrata Roy

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Education

University of California, Santa Cruz, USA **2016–2022**
PhD, Astronomy/Astrophysics; supervisor: Prof. Kevin Bundy
PhD defense: **June 2022**.
Thesis: *Star formation suppression and feedback in nearby galaxies.*

Presidency University, Kolkata, India **2014–2016**
M.Sc., Physics; thesis supervisor: Prof. Durgesh Tripathi (IUCAA)
Thesis: *Characteristics of Chromospheric Spectral Line in Solar Flares.*

Presidency University, Kolkata, India **2011–2014**
B.Sc., Physics; project supervisor: Prof. Ritaban Chatterjee
Project: *Probing the jets of blazars using the temporal symmetry of their multiwavelength outbursts*

Positions Held

Assistant Professor, Raman Research Institute, Bangalore, India **Aug 2026–**

Exploration Prize Postdoctoral Fellow, Arizona State University, USA **Aug 2025–Jun 2026**
(Independent fellowship; collaborators: S. Borthakur, R. Windhorst)

Postdoctoral Fellow, Johns Hopkins University, USA **Sep 2022–Jul 2025**
(Collaborators: T. Heckman, A. Henry.)

Other Positions

Visiting Scientist, Space Telescope Science Institute, USA **Sep 2022–Jul 2025**

Visiting Scientist, Center for Computational Astrophysics, Flatiron Institute, USA **Sep 2021**

Osterbrock Scholar, University of California, Santa Cruz **Sep 2019–Jun 2022**

Visiting Research Fellow, IUCAA, India **January 2016–May 2016**

Summer Research Fellow, Harish-Chandra Research Institute, India **May 2015–July 2015**

Honours & Awards

- **Exploration Prize Postdoctoral Fellowship**, Arizona State University (2025).
- **STScI Prize Fellowship** (2022; offer declined)
- **UC Dissertation Year Chancellor's Fellowship** (2021).
- **AAS National Osterbrock Leadership Program Fellowship** (2019).
- **Summer Research Fellowship**, Harish-Chandra Research Institute (2015).
- **INSPIRE Scholarship**, Department of Science & Technology, Government of India (2011–2016; awarded to students in the top $\sim 1\%$ pursuing basic sciences).

Invited Talks

18 invited talks among 47 presentations. Selected presentations can be found here.

- **Oct 2026** (scheduled): WAVES meeting, ICTS–TIFR – invited talk.
- **Sep 2026** (scheduled): Cosmic Connections: AI/ML Across Cosmic Scales, Institute of Mathematical Sciences – plenary talk.
- **Apr 2026**: NASA Galaxies Science Interest Group (SIG) seminar – invited seminar.
- **Mar 2026**: Cosmology Seminar, Arizona State University – invited seminar.
- **Feb 2026**: AGN & Energy Flows invite-only workshop, Flatiron CCA.
- **Jan 2026**: NASA AGN Science Interest Group (SIG) seminar – invited seminar.
- **Jun 2025**: JWST at the Gates of Cosmic Dawn, Johns Hopkins University – invited talk.
- **Mar 2024**: CTC Seminar, University of Maryland, College Park – invited colloquium.
- **Aug 2023**: Santa Cruz Galaxy Conference – invited talk.
- **Aug 2022**: Invited colloquium at School of Astrophysics, Presidency University.
- **Aug 2022**: Invited colloquium at Raman Research Institute.
- **Aug 2022**: Invited colloquium at Indian Institute of Astrophysics.
- **Aug 2022**: Invited colloquium at IISER Kolkata.
- **Jul 2022**: IISER Pune – invited colloquium.
- **Jun 2022**: SDSS-MaNGA Scicon3 – invited talk.
- **Sep 2021**: Harvard Center for Astrophysics – invited colloquium.
- **Aug 2020**: Presidency University Alumni Lecture Series – invited lecture.
- **Nov 2019**: University of California, Berkeley – invited talk.

Publications & Preprints

62 refereed papers; 11 first-author; 3 second-author; 1609 citations; h-index 23.
Student-led papers are marked †. Full list of publications: NASA ADS.

First- and second-author publications and preprints

1. **Roy, N.**, Borthakur, S., Heckman, T., Singh, T. (2026), "Lighting Up the CGM: Strong, Jet-Aligned Hα Emission around Radio Galaxies," arXiv:2606.30088; under review in ApJL.
2. **Roy, N.**, Heckman, T., Henry, A. (2026), "Mapping Jet-Gas Coupling and Energetic Ionized Outflows in High-Redshift Radio Galaxies with JWST/NIRSpec," ApJ, 1002, 57.
3. **Roy, N.**, Henry, A., et al. (2025), "MSA-3D: Uncovering Weak AGNs and Resolved Outflows in Disguise in $z \sim 1$ Star-Forming Galaxies," arXiv:2510.11326; under review in ApJ.
4. **Roy, N.**, Heckman, T., Henry, A., et al. (2025), "Lyman continuum leakage from massive leaky starbursts: A different class of emitters?" ApJ, 992, 91.
5. **Roy, N.**, Heckman, T., et al. (2024), "JWST Reveals Powerful Feedback from Radio Jets in a Massive Galaxy at $z = 4.1$," ApJ, 970, 69.
6. **Roy, N.**, Henry, A., Heckman, T., et al. (2023), "Early Results from GLASS-JWST XXII: Rest-frame UV-Optical Spectral Properties of Lyα Emitting Galaxies at $3 < z < 6$," ApJL, 952, L14.
7. **Roy, N.**, Moravec, E., Bundy, K., et al. (2021), "Radio Morphology of Red Geysers," ApJ, 922, 230.
8. **Roy, N.**, Bundy, K., et al. (2021), "Signatures of inflowing gas in red geyser galaxies hosting radio-AGN," ApJ, 919, 145.
9. **Roy, N.**, Bundy, K., et al. (2021), "Evidence of Wind Signatures in the Gas Velocity Profiles of Red Geysers," ApJ, 913, 33.
10. **Roy, N.**, Chatterjee, R., et al. (2019), "Probing the jets of blazars using the temporal symmetry of their multiwavelength outbursts," MNRAS, 482, 743.
11. **Roy, N.**, Bundy, K., et al. (2018), "Detecting Radio AGN Signatures in Red Geysers," ApJ, 869, 117.
12. † Moghni, A., **Roy, N.**, et al. (2026), "Galactic Rain: Cold Gas in Red Geyser Galaxies and Their Connection to Radio Activity and Interactions," ApJ, 1003, 31.
13. † Dasgupta, A., **Roy, N.**, Samui, S., et al. (2026), "Spatially Resolved Gas-Phase Metallicity Gradients in 382 MaNGA Low-Mass Galaxies: The Role of Concentration, Gas Velocity Dispersion, and Environment," under review in A&A.
14. Heckman, T., **Roy, N.**, et al. (2024), "Mergers, Radio Jets, and Quenching Star-Formation in Massive Galaxies," ApJ, 977, 125.
15. **Roy, N.**, Xu, X., Abdurro'uf, et al. (2025), "Feedback in Action at Cosmic Dawn: Spatially Resolved Galactic Outflows at $z = 6.2$," in preparation.

Selected co-authored publications

- Roberts-Borsani, G., et al. (incl. **Roy, N.**) (2023), "The nature of an ultra-faint galaxy in the cosmic dark ages seen with JWST," *Nature*, 618, 480.
- Borthakur, S., et al. (incl. **Roy, N.**) (2026), "Revealing Cosmic Ecosystems with the Hubble Space Telescope in the 2030s and Beyond," white paper submitted to the STScI Call for HST White Papers, arXiv:2606.11506.
- Boyett, K., et al. (incl. **Roy, N.**) (2024), "A massive interacting galaxy 510 million years after the Big Bang," *Nature Astronomy*, 8, 657.
- Kale, R., et al. (incl. **Roy, N.**) (2025), "Galaxies, Clusters, and Intergalactic Medium: A Vision Document for the Indian Astronomical Community," *Journal of Astrophysics and Astronomy*, 46, 32.
- Saxena, A., et al. (incl. **Roy, N.**) (2024), "Widespread AGN feedback in a forming brightest cluster galaxy at $z = 4.1$ unveiled by JWST," MNRAS, 531, 4391.
- Castellano, M., et al. (incl. **Roy, N.**) (2023), "Early Results from GLASS-JWST. XIX. A High Density of Bright Galaxies at $z \sim 10$ in the A2744 Region," ApJL, 948, L14.
- Bergamini, P., et al. (incl. **Roy, N.**) (2023), "The GLASS-JWST Early Release Science Program. III. Strong-lensing Model of Abell 2744 and Its Infalling Regions," ApJ, 952, 84.
- Mascia, S., et al. (incl. **Roy, N.**) (2023), "Closing in on the Sources of Cosmic Reionization: First Results from the GLASS-JWST Program," A&A, 672, A155.
- Xu, X., et al. (incl. **Roy, N.**) (2025), "Shining a Light on the Connections between Galactic Outflows Seen in Absorption and Emission Lines," ApJ, 984, 94.

Funding & Observing Awards

Total external support: approximately $1.1 million, including ∼ $430k from JWST, ∼ $220.5k from HST, ∼ $160k from NSF, and competitive fellowships.

Selected funding awards: Exploration Prize Postdoctoral Fellowship (∼ $255k); UC Chancellor's Dissertation Year Fellowship (∼ $52k); National Osterbrock Leadership Program Fellowship (∼ $15k); STScI Spring Symposium organization funds (∼ $10k).

Selected successful PI/deputy-PI observing proposals

- HST Cycle 33 GO-18096, *COS-RAD: Where Radio Jets Meet the Halo Gas* – PI; 26 orbits (2025).
- HST Cycle 31 GO-17485, *Dissecting Red Geyser Winds* – PI; 5 orbits (2023).
- LBT/MODS DDT, *Probing the quasar-radio galaxy pair* – PI; 0.5 night (2026).
- MMT/Binospec IFU DDT, *Where Radio Jets Meet the Halo Gas* – PI; 0.5 night (2026).
- Magellan/FIRE, *Quiet Galaxy, Large-scale Winds* – PI; 1 night (2025).
- GMRT 49-066, *Revealing Aged Radio Plasma and AGN Feedback in Red Geysers* – PI; 4 hr (2025).
- MMT/Binospec ID 331, *Jet–ISM coupling along the radio lobes* – PI; 2 nights (2025).
- LBT 2025A, *Exploring the Physics of a New Class of Lyman-Leaky Galaxies* – PI; 1 night (2025).
- Magellan/LLAMAS 2024A ID 54, *Mapping Outflows from Starbursts into the CGM* – deputy PI; 1 night (2024).
- Keck/OSIRIS 2022B-U051, *Probing feedback in low-luminosity AGN hosts* – science PI/deputy PI; 2 nights (2022).
- APEX P105-105.20EE, *Cold gas on the red sequence: maintenance mode feedback in action* – science PI/deputy PI (2020).

Selected co-I programs: JWST GO-10208 (115 hr), JWST GO-10815 (15 hr), JWST GO-7041 (78 hr), JWST GO-7076 (87 hr), JWST GO-5293 (10 hr), and JWST GO-6074 (8 hr); XRISM Cycle 1; HST archival programs AR-17859, 17559, and 17042, VLT/ERIS P116 (5 hr); Keck OSIRIS 2018B-U081.

Significant Contributions

- Provided some of the first multi-wavelength observational evidence for the theoretically-proposed "maintenance-mode" AGN feedback in quiescent galaxies through the discovery and characterization of **red geysers**. Led studies establishing their radio-AGN signatures, galaxy-scale winds, gas inflows, and connection between cold gas and radio activity, helping establish red geysers as an important laboratory for understanding how massive galaxies remain quenched.
- Demonstrated that powerful radio galaxies hosting relativistic jets can strongly disrupt, accelerate, and couple energy to ambient gas at high redshift. Used spatially resolved JWST/NIRSpec spectroscopy to map jet–gas coupling and energetic ionized outflows, including direct evidence for powerful radio-jet feedback in a spectacular massive galaxy at $z = 4.1$.
- Helped pioneer the **MSA-3D slit-stepping strategy** with JWST/NIRSpec, which systematically steps the Micro-Shutter Array across many galaxies to construct spatially resolved, IFU-like 3D spectroscopic data cubes for dozens of targets simultaneously. This multiplexed approach increases observing efficiency by a factor of ∼15, making large 3D spectroscopic surveys of galaxy populations feasible for the first time. This strategy can be implemented in other space-based missions like India's INSIST, and can be efficiently used to transform multi-object slits to a multi-object IFU.
- Built and led an extensive multi-wavelength observational program spanning HST, JWST, Keck, Magellan, MMT, LBT, VLT, GMRT, and ALMA. Secured access to more than **300 hours of HST and JWST observations**, served as PI or co-I on multiple (**40+hrs**) competitive ground-based telescope programs, and secured approximately **$1.1 million** in research support.
- Recognized for scientific leadership through invitations to shape future research priorities, for example India's Astronomy Vision Document, the Habitable Worlds Observatory AGN science case, and the white paper *Revealing Cosmic Ecosystems with the Hubble Space Telescope in the 2030s and Beyond*. Invited to serve on NSF grant-review, JWST and HST time-allocation, and TMT/WFOS' (early proposed instrument) planning and development team.
- Built a strong record of **research mentorship and capacity building**, mentoring nine students across high-school, undergraduate, Master's, and PhD levels. These projects have produced accepted and submitted student-led publications and helped mentees earn the prestigious Goldwater and Ron Ruby Scholarships, best-poster recognition, and best-speaker awards. One of the student-led papers was selected for a press release by AAS.
- Contributed to scientific community-building through the organization of conferences, colloquia, and professional-development programs, including co-creating the **CARINAS** monthly seminar series to increase the visibility of research by Indian women astronomers, organizing the STScI symposium *Recipes to Regulate Star Formation at All Scales*, and conducting multiple journal clubs, seminar series, etc.

Mentoring & Teaching

Mentored nine students: 3 high school, 3 Bachelor's, 1 Master's, and 2 PhD students.

Selected mentoring

- **Timothy Honablew**, Johns Hopkins PhD student (2024–present): outflows in nearby Seyfert galaxies; first of three planned papers in preparation; doctoral candidacy completed Feb 2026.
- **Arian Moghni**, UC Santa Cruz undergraduate and Rowland Fellow (2024–present): primary mentor on AGN feedback in low-luminosity AGN hosts. First student-led paper accepted; second in preparation. Awards include Goldwater and Ron Ruby Scholarships and best poster at the UCSC APS presentation.
- **Anuroop Dasgupta**, Presidency University Master's student (2023–2025): primary mentor on metallicity gradients in dwarf galaxies. Thesis submitted; student-led paper submitted; best speaker at West Bengal Regional Science Congress.
- **Annie Gao**, Johns Hopkins PhD student (2023–2024): academic and career mentoring in ISM physics and research pathways.
- **Anand Sharma**, IISER Pune undergraduate (2022): mentored project on AGNs in passive elliptical galaxies.
- **Atirath Dhara, Kaela McConnell, Jurij Waite**, UC Santa Cruz Science Internship Program (2017): primary mentor; work presented and published at the 2018 AAS meeting.
- **Marina Huang**, UC Santa Cruz undergraduate (2017): co-advisor on an automatic algorithm to identify MaNGA red geysers.

Teaching

- Guest instructor, LAMAT Summer Research Training Program, UC Santa Cruz (2021): Python workshops and hands-on astronomical data-analysis tutorials for ~ 10 students.
- Co-designed the introductory astronomy curriculum, Python modules, course structure, reading materials, and short-thesis ideas for Presidency University's new Astrophysics Master's program (2020–2023).
- Teaching Assistant, ASTR 5: *Overview of the Universe*, UC Santa Cruz (2018; $\sim$ 150 students): discussion sections, homework and worksheets, and tutorials.
- Teaching Assistant, ASTR 2: *The Formation and Evolution of the Universe*, UC Santa Cruz (2017; $\sim$ 150 students): office hours, grading, and tutorials.

Research Leadership & Service

- Science leadership in international collaborations: MSA-3D AGN/outflow science lead; GLASS-ERS data-analysis lead; Cosmic Springs/MACS0308 science lead on gas flows, resolved ionization, and metallicity; COS-RAD science PI and data-analysis lead.
- Invited review panelist: NSF Astronomy and Astrophysics Grants (2025), JWST TAC (2025), HST TAC (2024), and GMRT proposals (2025); manuscript referee for MNRAS and AAS journals (2020–present).
- Core group member, Habitable Worlds Observatory AGN Science Planning Committee (2024–2025); invited contributor to India's Astronomy Vision Document.
- Co-creator of the CARINAS monthly virtual seminar series highlighting research by Indian women astronomers; organizer of conferences, colloquia, and professional-development programs.

Scientific Community Building

- Organized the Beus Seminar Series at ASU (2025–present), the STScI AGN & Galaxies Journal Club and joint JHU–STScI colloquium (2024–2025), and JHU AstroCoffee (2023–2024).
- Organizer and SOC member for the 2024 STScI Spring Symposium (200+ participants); co-created the CARINAs Colloquium Series highlighting research led by Indian women astronomers.
- Organized Osterbrock leadership and career-development programs at UC Santa Cruz (2019–2021), including managing student mini-grants and the *Career Beyond Academia* panel.

Research Interests

Extragalactic astrophysics; galaxy evolution; black-hole and AGN feedback; baryon cycling in the ISM and CGM; star-formation quenching; relativistic jets; cosmic reionization and Lyman-continuum escape; multi-wavelength imaging and spatially resolved spectroscopy; astrostatistics and theoretical modeling.

Links

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